

Hisham Panamthodi Kajahussain

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Summary

Software Engineer with **4 years** building enterprise **Java/Spring Boot microservices** and **event-driven**, cloud-native **AWS systems** across healthcare and life sciences. Delivered measurable impact via **process automation**, **API optimization**, and **cloud-native architecture**, with hands-on experience shipping production-grade **AI/ML applications**.

Education

California State University (CSUF), Fullerton, CA

Aug 2024 – May 2026

M.S. in Computer Science

GPA: 3.9/4

Relevant Coursework: Data Structures and Algorithms, Machine Learning(ML), Software Design and Architecture, Big Data

APJ Abdul Kalam Technological University (KTU), Kerala, India

Jul 2015 – May 2019

B.Tech in Computer Science and Engineering

GPA: 3.81/4

Professional Experience

Tata Consultancy Services (TCS)

Cochin, India

Software Engineer | Java, Spring Boot, Microservices, REST APIs, AWS, Kubernetes, PostgreSQL

July 2019 – July 2023

- Improved Bayer AG’s campaign operational efficiency by **35%** by developing a **Java/Spring Boot** middleware application that automated end-to-end campaign workflows including campaign targeting, campaign simulation, target file generation, and **Salesforce Marketing Cloud** campaign delivery across **4 countries**.
- Implemented event-driven simulation pipeline using **RabbitMQ**, **Spring WebClient**, and **WebSocket** to decouple compute-intensive Sales Rep-to-HCP mapping from synchronous HTTP threads, reducing campaign simulation turnaround time by **60%** with real-time frontend notifications.
- Eliminated hardcoded credentials across **7 microservices** using **AWS Secrets Manager** and **IAM Assume Roles**; contributed to **ECS Blue-Green CI/CD** enabling **zero-downtime releases** across **4 production environments**.
- Developed **PostgreSQL** data access layers using **Spring Data JPA** and **Hibernate** for multi-stage campaign persistence, and implemented **Spring AOP** audit logging to enforce end-to-end traceability, reducing root-cause analysis time by **30%**.
- Engineered **Java/Spring Boot** microservices for Bayer AG’s Crop Science platform monitoring pesticide, herbicide, and fungicide resistance in agricultural field samples, supporting sample-to-report workflows for **10,000+ samples annually**.
- Decomposed a legacy **Oracle-based monolith** into **7 domain-driven microservices** on **AWS EKS**, cutting release cycle time by **60%**; implemented a **7-state sample lifecycle** and **Amazon SES** alerts for state changes.
- Reduced API response times by **45%** by introducing **Redis ElastiCache** TTL caching and optimizing **MySQL** reads for high-frequency reference data under concurrent load.
- Secured **90+ REST API endpoints** across **7 microservices** using **Spring Security 5**, **JWT RS256**, and **AD/LDAP**, enforcing **4-tier RBAC** via centralized **API Gateway** validation.

Skills

Languages & Frameworks: Java, Spring Boot, JPA/Hibernate, Python, FastAPI, JavaScript, React, Node.js, HTML/CSS

Cloud Platforms: AWS (EC2/S3/RDS/ECS/EKS/Lambda/IAM/SNS/SES/ElastiCache/Secrets Manager)

DevOps & Tools: Docker, Kubernetes, Terraform IaC, Git/GitHub Actions, CI/CD, Kafka, RabbitMQ, JUnit/Mockito

Databases: MySQL, PostgreSQL, MongoDB (NoSQL), Redis, ChromaDB

Architecture: Microservices, Event-Driven Architecture, Distributed Systems, API Gateway, Blue-Green Deployment, Agile

Academic Projects

AI Clinical Decision Support System | GitHub

Jan 2026 – May 2026

- Built an end-to-end **clinical AI platform** using **Java/Spring Boot**, **4 Python/FastAPI microservices**, and **LLaMA 3.2 RAG**, transforming unstructured patient notes into ranked diagnoses, treatment recommendations, and explainable AI insights — reducing clinician decision time from **~30 minutes** to under **1 minute per case**.
- Fine-tuned **Bio ClinicalBERT** on **1M+ ICU clinical notes (MIMIC-III)** achieving **87.45% accuracy** across **24 classes**; deployed on **Amazon Web Services (AWS) ECS Fargate** via **Terraform IaC** and **GitHub Actions Continuous Integration/Continuous Deployment (CI/CD)**, reducing deployment time from hours to under **10 minutes**.

CloudShelf E-Commerce Platform | GitHub

Aug 2025 – Dec 2025

- Built a cloud-native bookstore platform on **AWS EKS** with **9 independently deployable microservices** and polyglot persistence (PostgreSQL, MongoDB, Redis) across **3 AWS Availability Zones**, eliminating any single point of failure.
- Built an async order pipeline using **Apache Kafka** and **AWS SNS**, cutting confirmation latency from **~4s** to under **500ms** and ensuring delivery during downstream outages.
- Integrated **Resilience4j Circuit Breaker** across **9 microservices** to prevent **cascading failures**; automated **CI/CD** via **GitHub Actions** and **AWS ECR**, reducing manual release effort by **~80%**.

Publications

- Hisham P K, “*MedIntelliX – AI-Powered Medical Decision Support System*,” **IEEE CHASE 2026 (top 18% selectivity)**.